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Expanding the Asteraceous flora: New records from Gorakhpur District, Uttar Pradesh

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Abstract- This paper reports the addition of four species belonging to three genera and three tribes of family Asteraceae to the flora of Gorakhpur district. It provides a brief description of each taxon, including phenological data, distribution, ecology, and voucher specimen number. Photographs of each taxon are also included.

Keywords: Addition, Diversity, Terai region, Asteraceae, Gorakhpur

INTRODUCTION

Gorakhpur district is located in the northeastern terai region of Uttar Pradesh, India. It extends between Latitudes 26°52 N and 27°292 N and Longitudes 83°042 E and 84°262 E.[1] The district is situated within the basins of the Rapti and Rohini rivers. It shares borders with the Maharajganj district to the north, the Deoria district to the east, the Sant Kabir Nagar district to the west, and the Mau District to the south.

The floristic diversity of Gorakhpur district has been studied by numerous taxonomists over the years.[1-17] Their work indicates that the species listed in this paper have not been previously reported from Gorakhpur district. During the floristic survey of the family Asteraceae in Gorakhpur, from November 2019 to March 2024, the author identified four species belonging to three genera and three tribes of family Asteraceae, which are new additions to the flora of Gorakhpur. Each specimen has been accurately cited with updated information, including type detail a brief description, phenology, chromosome number, distribution, ecological notes, uses, and specimen examined.

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MATERIALS & METHODS

Extensive field surveys were conducted from November 2019 to March 2024 to investigate the diversity of the family Asteraceae in Gorakhpur district. During their flowering period, plant specimens were collected and herbaria were prepared according to standard procedures. Identification was carried out using relevant literature, including published books and floras.[2,18-20] Distribution data, locality information, GPS coordinates, and photographs of live plant samples were recorded. Voucher specimens were prepared in accordance with Jain & Rao (1977)[21] and deposited at the DUTHIE Herbarium, Department of Botany, University of Allahabad, and Department of Botany, DDU Gorakhpur University, Gorakhpur. Additionally, herbarium references from CAL, BSA, DD, BSD, LWG, PRFH, and GPU were consulted for the correct identification. The taxonomy of newly added taxa was verified using credible online databases like IPNI, Tropicos,

Plants of the World Online (POWO), and World Flora Online (WFO).

ENUMERATION OF THE SPECIES

TRIBE 1: HELIANTHEAE

The genus *Acmella* Rich. ex Pers. was first described by Richard in (1807)[22] & Rahman *et al.*, (2016)[23] It belongs to the tribe Heliantheae and subtribe Spilanthinae of the family Asteraceae, which comprises 36 species worldwide and 11 taxa (ten species and one variety) in India.[24]Agnihotri *et al.* (2024)[25] reported five species of Acmella in Uttar Pradesh, of which only *Acmella oleracea* was reported from Gorakhpur and adjoining regions[26]. It is globally distributed in tropical and subtropical regions extending to Central USA.[27]

Types- LECTOTYPE, here designated: *Acmella oppositifolia* (Lamarck) R.K. Jansen[28].

Key to the species

1a. Capitula arranged in panicle…… **(3)** *A. paniculata*

1b. Capitula not arranged in panicle… **(2)** *A. ciliata*

Taxonomic treatment

1. *Acmella ciliata* (Kunth) Cass., G.-F. Cuvier, Dict. Sci. Nat. ed. 2. 24: 331. 1822; *Spilanthes ciliata* Kunth in F.W.H. von Humboldt, A.J.A. Bonpland & C.S. Kunth, Nov. Gen. Sp. Pl. ed. fol., 4:163. 1818; H.B.K., Nov. Gen. et Sp. PI. 208. 1820; H.J. Chowdhary in Hajra *et al*., Fl. India 12: 409"410. 1995.

Type- COLOMBIA. Near Chipo and Santa Fe de Bogota, [Jun-Sep. Fide Sprauge (1968)] 1801, Bonpland s.n. (holotype: P-HBK, IDC 6209. 107: III.5, photo: OS!)[28].

A perennial herb with rooting at lower nodes, 40 50 cm tall. The stem is decumbent, slender, solid, pubescent, and reddish-green. Leaves are 3 4×2 2.5 cm, simple, petiolate, opposite, ovate to broadly ovate with an acute apex and a truncate base, with serrate margins and a rough lamina. Capitula are solitary or arranged in 2 3 axillary or terminal clusters; they are heterogamous, radiate, and 9 10 mm in diameter. The phyllaries are numerous, arranged in 2 series; the outer bract is green, elliptic-lanceolate with an obtuse apex, and varies from glabrous to sparsely pubescent, measuring 3 6 × 1 2 mm. Ray florets number 9 10 per capitulum are yellow, 4 5 lobed, pubescent, and 5 mm long. Disc florets are tubular, yellow-orange, 5-lobed, and 5 mm in length. Cypselae are obovate, black, with ciliated margins, corky texture, and absent ribs, measuring 3 4 × 2 mm; sometimes, the pappus is absent; occasionally, there are two very short white bristles about 0.5 mm.

Common Name: Fringed Pod, Toothache plant, Eyeball plant

Flowering and fruiting: September-December

Distribution and ecological notes: This species is abundant in the region, residing near forests, woodlands, savannas, shrublands, wetlands, and artificial terrestrial environments (Fig. 1). It is commonly found near waste sites. Native to countries including Argentina, Bolivia, Brazil, Colombia, Ecuador, Guyana, Panamá, Peru, and Venezuela.[27] It has been introduced to areas such as India, Malaya, Sulawesi, Sumatra, Taiwan, and Thailand.

Uses: Used in anti-ageing herbal tonics and mouthwashes.[29,30]

Specimen examined: UTTARAKHAND; Dehradun (FRI Campus), 15 Nov 2020, *Jeetendra Kumar Vaishya & A. A. Ansari* 92281, 92279, 92278 (BSA!). ASSAM; Dina Hasao, 27 Oct 2017, *Rohit Kr. Verma* 312372 (LWG!). **UTTAR PRADESH;** Meerut (C.C.U Campus), 15 Nov 2011, S.n. 92289 (BSA!); Near Gurudwara, 25 Aug 2011; *Jeetendra Kumar Vaishya & A.A. Ansari* 92285, 92285,92288 (BSA!); Baghpat (roadside), 16 Sep 2012; *Jeetendra Kumar Vaishya & A.A. Ansari* 92282, 92283 (BSA!); Kishanpur Wildlife Sanctuary, 24 Dec 2015, *P. Katiyar* 304046 (LWG!); Nawabganj Bird Sanctuary, 27 Nov 2011, *B.K. Shukla & Arti Garg* 88741, 88742 (BSA!); Gonda (Parvati Arya), 05 Jan 2007, *Dr. K. K. Khanna* 83310, 83311, 83316 (BSA!); Basti (Badeban), 18 Jan 2007, *Dr. K.K. Khanna* 83634, 83635 (BSA!); Allahabad (Railway Station All Junction), 17 April 2011; *Jeetendra Kumar Vaishya & A. A. Ansari* 92288, 92288,92287 (BSA!); Allahabad (B.S.I Garden), 26 March 2011; *Jeetendra Kumar Vaishya & A. A. Ansari* 92288 (BSA!); Gorakhpur (D.D.U. Gorakhpur University Campus), 15 March 2020, elevation 58m; Lat. N 26°45'00.61'' & Long.E 083°02'56.27''; *Preeti Gupta, S. Shekhar & V. K. Madhukar* PG900107 (GPU!); Jungle Dhusan, 18 March 2022, *Preeti Gupta, S. Shekhar & V. K. Madhukar* PG900109 (GPU!); Unawal-Mishraulia, 13 Feb 2022, *Preeti Gupta, S. Shekhar & V. K. Madhukar* PG900108 (GPU!).

2. *Acmella paniculata* (Wall. Ex DC.) R.K. Jansen, Syst. Bot. Monogr, 8: 67. 1985; B.K. Shukla in Sinha & Shukla (Eds). Fl. Uttar Pradesh 2: 34. 2020. *Spilanthes paniculata* Wall. Ex DC., Prodr. 5: 625. 1836; H.J. Chowdhery in Hajra *et al*., Fl. India 12: 410 412. 1995. *Spilanthes acmella* var. *paniculata* (DC.) C.B. Clarke, Comp. Ind. 139.1876; Hook.f., Fl. Brit. India 3: 307. 1881.

Spilanthes calva DC. in Wight, Contrib. Bot. Ind. 19. 1834; *S. acmella* var. *calva* (DC.) C.B. Clarke, Comp. Ind. 138. 1876.

Type- INDIA. Mantaban, 1827, Wallich 296(holotype: G-DC, IDC 800. 963: II.5)[28].

This annual herb reaches 40-50 cm in height. Its stem is erect, branched, slender, hollow, pubescent, and reddish-brown. The leaves measure 1.5-8×1-6 cm, are simple with a 4 mm petiole, opposite, ovate-elliptic to elliptical with an acute apex and an attenuate base, three-nerved, with serrate margins, and a pubescent lower surface. The capitula are arranged in a panicled, heterogamous, radiate, conical cluster, 6 7 mm in diameter; they sit on peduncles up to 10 cm long. The numerous phyllaries are two-seriate; outer bracts are green, elliptic-lanceolate to ovate with an obtuse apex, either glabrous or sparsely pubescent, measuring 3–5×1–2 mm. Each capitulum contains 5 6 yellow ray florets with three lobes, and yellow-orange disc florets with five lobes. The cypselae are obovoid, black, 2–3×2 mm, ciliated with a corky margin, lacking ribs; pappus is absent or consists of two 0.5 mm long white awns.

Common Name: Panicled spot flower

Flowering and fruiting: October-December

Distribution and ecological notes: Found near waste sites (Fig. 1). It is an annual that mainly grows during the seasonally dry tropical biome. The native range of this species includes China, Taiwan etc. It has been introduced into Colombia, Ecuador, Pakistan, and Panamá.[27]

Uses: The leaves are traditionally used to alleviate toothache and address infections of the throat and gums. It is also used as a poison and food. Additionally, they help combat bacterial and fungal skin conditions.[31,32]

Specimen examined: **UTTARAKHAND**; Haldwani, 17 Feb 1988, *Jeetendra K. Vaishya & A.A. Aansari* 95654 (BSD!); **UTTAR PRADESH**; Upper Ganga Ramsar site (Ahar Ghat), 24 September 2012, *A. Garg & V.K. Singh* 100915, 100914 (BSA!); Anand Nagar, 15 Jan 2003, *S.K. Singh* 4112 (GPU!); Shravasti, 14 March 2008, *Dr. K.K. Khanna* 86456 (BSA!); Sant Kabir Nagar (Bakhira), 21 Oct 2008, *K.K. Khanna* 82173 (BSA!); Gorakhpur (Sardarnagar), 8 Nov 2022, elevation 69 m; lat. N 26°36'53.72" & long. E 83°34'31.10", *Preeti Gupta, S. Shekhar & V. K. Madhukar* PG9001041 (GPU!); Gorakhpur (Chauri-Chaura), 17 Nov 2022, *Preeti Gupta, S. Shekhar & V. K. Madhukar* PG9001042 (GPU!).



Fig. 1- Habit: A. *Acmella ciliata* (Kunth) Cass.; **B.** *Acmella paniculata* (Wall. Ex DC.) R.K. Jansen; **C.** *Blumea obliqua* (L.) Druce; **D.** *Erigeron sublyratus* Roxb. Ex. DC [*Based on: A-PG90017 (GPU!); B- PG9001041 (GPU!); C- PG9001036 (GPU!); D- PG900130 (GPU!)*]

TRIBE 2: INULEAE

The genus *Blumea* DC. was first described by A.P. de Candolle (1836)[33] & (Paul & Devi, 2014)[34]. It belongs to the tribe Inuleae and sub-tribe Pluchenieae within the Asteraceae family.[35,36] This genus comprises about 97 species found in tropical and subtropical regions of Southeast Asia, Africa, and Australia.[27] In India, 29 species have been documented.[35] Specifically, in Uttar Pradesh, 12 species are recorded[20,25] with 6 of these species reported from Gorakhpur.[10, 11,26]

Type- *Blumea balsamifera* (L.) DC. (Typ. Cons.)[37].

Taxonomic treatment

3. *Blumea obliqua* (L.) Druce, Rep. Bot. Exch. Club Brit. Iles 4 (Suppl.2):609. 1916 publ.1917; S. Kumar in Hajra *et al.*, Fl. India 13: 137. 1995; B.K. Shukla in Sinha & Shukla (Eds). Fl. Uttar Pradesh 2: 48. 2020. *Erigeron obliquum* L., Mant. Pl. 2: 573. 1771. *Blumea amplectens* DC., R. Wight, Contr. Bot. India 13. 1834; Hook.f., Fl. Brit. India 3: 260. 1881; Duthie, Fl. Gangetic Plain 1: 451–452. 1903. *Blumea pubiflora* DC, Prodr. 5:434. 1836.

Type- Lectotype: Herb. Linn. No. 994.1 (LINN!)[38].

Annual herb standing 20–40 cm tall, with an erect, sparsely branched stem that is slender, solid, and hairy. The stem is yellow to brownish-green. Leaves measure

4–5 × 2–3 cm, are simple, sessile, and alternate. They are pubescent, elliptic to oblong–lanceolate, with an acute apex and a half-amplexicaul base, featuring serrated margins. The solitary, terminal, or axillary capitula are heterogamous and disciform, measuring 10–15 mm in diameter, on peduncles. They have pinkish-purple flowers; phyllaries arranged in 2–3 series. The outer bract is 1–8 mm, green with a purple tip, linear with an acuminate apex. The disc florets are tubular, bisexual, 5–6 mm long, yellow with purple tips, and have 5 lobes. Female florets have a 3–4 mm filiform corolla with 2–3 lobes. The oblong cypselae are pubescent, dark brown, 7×1 mm, ribless, and have no pappus ribs. The pappus is white, with 10–12 bristles measuring about 6 mm.

Common Name: Oblique Blumea

Flowering and Fruiting: February-June

Distribution and ecological notes: The native range of this species extends from the Indian Subcontinent to Indo-China and Java. It is a subshrub mainly growing in seasonally dry tropical biomes, often in moist, shady roadside areas. Its distribution includes the Andaman Islands, Assam, India, Jawa, Laccadive Islands, Myanmar, Pakistan, and Sri Lanka.[27]

Uses: Used as a remedy for malaria, influenza, bronchitis, and asthma.[39]

Specimen examined: RAJASTHAN; Nagkanda (Lohargal), 03 March 1960, *N.C. Nair* 17812 (BSD!). **UTTAR PRADESH;** Upper Ganga Ramsar Site (Khadar Road), 22 Feb 2016, *Dr. Arti Garg* 100601 (BSA!); Daulatabad, 13 Oct 1972, *Prithipal Singh* 27108 (BSD!); Allahabad, 27 April 1967, *G. Panigrahi* 18161 (BSA!); Gorakhpur (Jungle Dhusan), 19 Feb 2022, elevation 44m; lat. N 26°47'45.47'' & long. E 83°25'52.66''; *Preeti Gupta, S. Shekhar & V.K. Madhukar* PG9001036 (GPU!); Pipraich, 17 April 2021, *Preeti Gupta, S. Shekhar & V.K. Madhukar* PG9001035 (GPU!). **MADHYA PRADESH;** Shadol (Charada Coalfield), 19 March 2006, *B. K. Shukla* 88893, 88895 (BSA!).

TRIBE 3: ASTEREAE

The genus *Erigeron* L. was first described by Carl Linnaeus in 1753.[40] It belongs to the tribe Astereae and subtribe Conyzinae, comprising 457 species mainly in temperate zones. It has a global distribution, with a strong presence in North America.[27] In India, about 18 genera and 83 species are recorded[19], including 5 in Uttar Pradesh[20,25] and 2 in Gorakhpur[11,1].

Type-*Erigeron uniflorus* L. (M.L. Green, Prop. Brit. Bot. 181.1929: W. Huber, Taxon 41: 563.1992-Herb. Linn. 994.23, typ. Conserve. Prop.)[41]

Taxonomic treatment

4. *Erigeron sublyratus* Roxb. ex DC., R. Wight, Contr. Bot. India 9. 1834; Hajra in Hajra *et al.*, Fl. India 12: 125. 1995; B.K. Shukla in Sinha & Shukla (Eds). Fl. Uttar Pradesh 2: 63. 2020.

E. asteroid Roxb., Fl. Ind. ed. 2,3: 432. 1832, nom.illeg.; Hook.f., Fl. Brit. India 3: 254. 1881; Duthie, Fl. Gangetic Plain 1: 447. 1903.

Type- INDIA. In arenosis et apricis provinciae Tanjorem, Wight 1403. P00691950 (Syntype!).[42]

Annual herb, 30 60 cm tall. Stem erect or decumbent, branched, slender, solid, pubescent, green. Leaves 5 6×2 cm, simple, sessile, alternate, pubescent, oblanceolate to spathulate with an obtuse apex, auriculate and semi-amplexicaul base, and serrate margins. Capitula either sub-corymbose or solitary on axillary peduncles, heterogamous, radiate, 6 8 mm in diameter; phyllaries 2-seriate, 5 mm long; outer bract green, pubescent, linear-lanceolate with an acuminate apex, and entire margins. Ray florets pinkish-purple, approximately 3 mm long, unlobed, and eligulate. Disc florets yellow, about 4 mm long. Cypselae obovoid, smooth, light yellow, without ribs, measuring 6 7 × 1 mm; pappus hairy, dirty white, terminal, approximately 4 mm long.

Common Name: Sonasali, White- flowered Fleabane

Flowering and Fruiting: April-August

Distribution and ecological notes: Grows in open fields. Native to the Indian subcontinent, Indo-China, Myanmar, Nepal, Sri Lanka, Thailand, and Vietnam.[27]

Uses: Wild, edible Shoots are used as Snacks and vegetables.[43]

Specimen examined: UTTAR PRADESH; Gorakhpur (Deoria bypass), 8 May 2022, *Preeti Gupta, S. Shekhar & V.K. Madhukar* PG900131 (GPU!); Turra nala, 7 April 2022, elevation 61 m; lat. N 26°45'12.77'' & long. E 083°29'10.38''; *Preeti Gupta, S. Shekhar & V.K. Madhukar* PG900130 (GPU!); Jungle Dhusan, 22 Aug 2022, *Preeti Gupta, S. Shekhar & V.K. Madhukar* PG900132 (GPU!). **MADHYA PRADESH**; AABR (along Narmada River, Amarkantak), 13 April 2005, S. L. Bondya, A. N. *Shukla, Smita Mishra* 74617, 74018 (BSA!). **WEST BENGAL**; Jaldapara National Park, 13 May 2014, *K. Kavhigeyan* 65039 (CAL!).

RESULT & DISCUSSION

The explorations, conducted from November 2019 to March 2024, led to the collection of several significant Asteraceae specimens from Gorakhpur district in Uttar Pradesh. After careful examination, observations were recorded and cross-checked with various floras. A thorough review of recent taxonomic publications and floras of Gorakhpur, along with the Flora of Uttar Pradesh (Vol. II), shows that these specimens are new additions to the district's Asteraceae flora. As a result, four species from three tribes and three genera within the Asteraceae family have been added to the region's flora.

CONCLUSION

This study reports four new taxa in the Asteraceae family, covering three tribes, three genera, and four species from Gorakhpur district, Uttar Pradesh. These results are important for updating the area's flora and offer essential insights into the region's plant diversity.

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